

Conductors in a magnetic field — Solution

X24 (2024) — English translation

A1

0.60 pts

Let the moment of time $t_0 = 0$ the load be at the origin of coordinates, and the projection of its velocity onto the axis x is equal to v_0 . Determine the dependences of the coordinate $x(t)$ and speed $v_x(t)$ of the load on time t . Express your answer using v_0 , γ , ω_0 and t .

Let's write down the equation of motion of the load:

$$m\ddot{x} = -\beta\dot{x} - kx \Rightarrow \ddot{x} + \frac{\beta}{m}\dot{x} + \frac{k}{m}x = 0.$$

Taking into account introduce notation:

$$\ddot{x} + 2\gamma\dot{x} + \omega_0^2 x = 0.$$

We seek solution in complex form:

$$x = \operatorname{Re} \{ A e^{\lambda t} \},$$

where $A \neq 0$ and λ – some complex number. Then:

$$A(\lambda^2 + 2\gamma\lambda + \omega_0^2) = 0 \Rightarrow \lambda = -\gamma \pm i\sqrt{\omega_0^2 - \gamma^2}.$$

Solution represent itself sum solution, corresponding different value λ :

$$x(t) = e^{-\gamma t} \operatorname{Re} \{ A_1 e^{i\sqrt{\omega_0^2 - \gamma^2}t} + A_2 e^{-i\sqrt{\omega_0^2 - \gamma^2}t} \} = C e^{-\gamma t} \sin \left(\sqrt{\omega_0^2 - \gamma^2}t + \varphi_0 \right),$$

where C and φ_0 are real numbers. Differentiate:

$$v_x(t) = C e^{-\gamma t} \left(\sqrt{\omega_0^2 - \gamma^2} \cos \left(\sqrt{\omega_0^2 - \gamma^2}t + \varphi_0 \right) - \gamma \sin \left(\sqrt{\omega_0^2 - \gamma^2}t + \varphi_0 \right) \right).$$

For moment time $t = 0$ have:

$$\begin{cases} x(0) = C \sin \varphi_0 \\ v_x(0) = C \left(\sqrt{\omega_0^2 - \gamma^2} \cos \varphi_0 - \gamma \sin \varphi_0 \right) \end{cases}$$

Taking into account initial condition:

$$\begin{cases} x(0) = 0 \\ v_x(0) = v_0 \end{cases} \Rightarrow \begin{cases} \varphi_0 = 0 \\ C = \frac{v_0}{\sqrt{\omega_0^2 - \gamma^2}} \end{cases}$$

Final we get:

Answer:

$$x(t) = \frac{v_0}{\sqrt{\omega_0^2 - \gamma^2}} e^{-\gamma t} \sin \left(\sqrt{\omega_0^2 - \gamma^2}t \right).$$

A2

0.40 pts

Obtain an exact expression for Q . Express your answer in terms of ω_0 and γ .

Let's rewrite the expression for the quality factor in the following form:

$$Q = \frac{2\pi}{1 - \left(\frac{v_1}{v_0}\right)^2},$$

where v_1 – the projection of the weight velocity at the second passage of the origin with the more same direction velocity. Quantity velocity v_1 be achieved through period T , equal:

$$T = \frac{2\pi}{\sqrt{\omega_0^2 - \gamma^2}}.$$

Then:

$$\frac{v_1}{v_0} = e^{-\gamma T} = e^{-2\pi\gamma/\sqrt{\omega_0^2 - \gamma^2}},$$

from where we get:

Answer:

$$Q = \frac{2\pi}{1 - e^{-4\pi\gamma/\sqrt{\omega_0^2 - \gamma^2}}}.$$

A3

0.20 pts

Derive an approximate expression for the quality factor Q at weak attenuation ($\gamma \ll \omega_0$). Express your answer using m , k and β .

The expansion of the denominator at $\gamma \ll \omega_0$ is as follows:

$$1 - e^{-4\pi\gamma/\sqrt{\omega_0^2 - \gamma^2}} \approx 1 - e^{-4\pi\gamma/\omega_0} \approx 1 - \left(1 - \frac{4\pi\gamma}{\omega_0}\right) = \frac{4\pi\gamma}{\omega_0},$$

from which obtain approximation for the quality factor for weak damping Q :

$$Q \approx \frac{\omega_0}{2\gamma}.$$

Substitute ω_0 and γ , we obtain:

Answer:

$$Q \approx \frac{\sqrt{mk}}{\beta(H)}.$$

B1

0.60 pts

The deviation x of the load from the position depends on time t as follows:

$$x(t) = A \sin(\Omega t + \varphi_0)$$

Find A and φ_0 . Answer express through A_0 , Ω , ω_0 and γ .

The equation of motion is as follows:

$$m\ddot{x} = k(A_0 \sin \Omega t - x) - \beta\dot{x},$$

from which:

$$\ddot{x} + 2\gamma\dot{x} + \omega_0^2 x = \omega_0^2 A_0 \sin \Omega t.$$

We seek solution in form:

$$x = \operatorname{Re} \left\{ \hat{A} e^{i\Omega t} \right\},$$

where $\hat{A} \neq 0$ – some complex number. Then:

$$\hat{A} ((\omega_0^2 - \Omega^2) + 2i\Omega\gamma) = \omega_0^2 A_0 e^{-i\pi/2}.$$

Express A :

$$\hat{A} = \frac{\omega_0^2 A_0 ((\omega_0^2 - \Omega^2) - 2i\Omega\gamma) e^{-i\pi/2}}{((\omega_0^2 - \Omega^2)^2 + 4\gamma^2\Omega^2)} = \frac{A_0 \omega_0^2 e^{-i(\pi/2 - \varphi)}}{\sqrt{(\omega_0^2 - \Omega^2)^2 + 4\gamma^2\Omega^2}},$$

where φ – argument complex number $z = (\omega_0^2 - \Omega^2) - 2i\Omega\gamma$. Extracting the real part, we find:

$$\Delta x(t) = \frac{A_0 \omega_0^2}{\sqrt{(\omega_0^2 - \Omega^2)^2 + 4\gamma^2\Omega^2}} \sin(\Omega t + \varphi).$$

Thus:

$$A = \frac{A_0 \omega_0^2}{\sqrt{(\omega_0^2 - \Omega^2)^2 + 4\gamma^2\Omega^2}} \quad \varphi_0 = \varphi.$$

For determining φ there are three cases:

$$\varphi = \begin{cases} -\arctan \frac{2\gamma\Omega}{\omega_0^2 - \Omega^2} & \text{for } \Omega < \omega_0 \\ -\frac{\pi}{2} & \text{for } \Omega = \omega_0 \\ -\pi - \arctan \frac{2\gamma\Omega}{\omega_0^2 - \Omega^2} & \text{for } \Omega > \omega_0 \end{cases}$$

So way:

Answer:

$$A = \frac{A_0 \omega_0^2}{\sqrt{(\omega_0^2 - \Omega^2)^2 + 4\gamma^2\Omega^2}},$$

$$\varphi_0 = \begin{cases} -\arctan \frac{2\gamma\Omega}{\omega_0^2 - \Omega^2} & \text{for } \Omega < \omega_0 \\ -\frac{\pi}{2} & \text{for } \Omega = \omega_0 \\ -\pi - \arctan \frac{2\gamma\Omega}{\omega_0^2 - \Omega^2} & \text{for } \Omega > \omega_0 \end{cases}$$

B2

0.30 pts

Obtain exact expressions for the resonant cyclic frequency Ω_{res} and the corresponding oscillation amplitude A_{res} . Express your answers using ω_0 , γ and A_0 . Consider that $\gamma\sqrt{2} < \omega_0$.

Differentiating the denominator with respect to Ω^2 , we obtain:

$$-2(\omega_0^2 - \Omega^2) + 4\gamma^2 = 0,$$

from where:

Substituting Ω_{res} into the expression for A , we find:

Answer:

$$\Omega_{res} = \sqrt{\omega_0^2 - 2\gamma^2}.$$

B3

0.30 pts

Derive approximate expressions for Ω_{res} , A_{res} and $\Delta\omega$ under weak damping ($\gamma \ll \omega_0$). Express your answers using A_0 , ω_0 and γ .

Simplified expressions for Ω_{res} and A_{res} are obtained trivially and take the following form:

Consider the cyclic frequency $\Omega = \Omega_{\text{res}} + \Delta\Omega$, where $\Delta\Omega \ll \Omega_{\text{res}}$. The value of Ω_{res} can be considered equal to ω_0 , since:

$$\Omega_{\text{res}} = \sqrt{\omega_0^2 - 2\gamma^2} \approx \omega_0 - \frac{\gamma^2}{\omega_0}.$$

Deviation Ω_{res} from ω_0 represent itself quantity second order smallness. Then for under the radical expression obtain:

$$(\omega_0^2 - \Omega^2)^2 + 4\gamma^2\Omega^2 \approx (\omega_0^2 - (\omega_0 + \Delta\Omega))^2 + 4\gamma^2(\omega_0 + \Delta\Omega)^2 \approx 4\omega_0^2\Delta\Omega^2 + 4\gamma^2\omega_0^2.$$

The quantity $\Delta\Omega$ is such that under the radical the expression is twice larger corresponding resonance. Hence:

$$4\Delta\Omega^2\omega_0^2 + 4\gamma^2\omega_0^2 = 8\gamma^2\omega_0^2 \Rightarrow \Delta\Omega_{1,2} = \pm\gamma.$$

Since $\Delta\omega = \Delta\Omega_1 - \Delta\Omega_2$, we get:

Answer:

$$\Omega_{\text{res}} \approx \omega_0 \quad A_{\text{res}} \approx \frac{A_0\omega_0}{2\gamma}.$$

C1

0.30 pts

Find the induction B_x of the magnetic field of the ring on its axis at the point with coordinate x . Express your answer in terms of x , R , I and the magnetic constant μ_0 .

From the Biot-Savart-Laplace law:

$$d\vec{B} = \frac{\mu_0}{4\pi} \frac{[\vec{r} \times d\vec{r}]}{r^3},$$

where $\vec{r}$ – radius-vector element ring relatively point with coordinate x . For each point ring:

$$r = \sqrt{R^2 + x^2}.$$

Integral by circuit, from the vector product we obtain from its geometric meaning:

$$\oint_L [\vec{r} \times d\vec{r}] = 2\vec{S} = 2\pi R^2 \vec{e}_x,$$

from:

Answer:

$$B_x(x) = \frac{\mu_0 I R^2}{2(R^2 + x^2)^{3/2}}.$$

C2

1.00 pts

Determine the magnetic moment $\vec{m}$ of the disk. Express your answer in terms of $\vec{e}_x$, r_0 , h , ρ and $\dot{B}$.

From Faraday's law of electromagnetic induction we obtain:

$$\mathcal{E}_{\text{ind}} = -\frac{d\Phi}{dt} = -\pi r^2 \dot{B} = 2\pi r E_{\text{vort}},$$

from which:

$$E_{\text{vort}} = -\frac{\dot{B}r}{2}.$$

From Ohm's law in differential form:

$$\vec{j} = \frac{\vec{E}}{\rho},$$

from which:

$$j(r) = -\frac{\dot{B}r}{2\rho}.$$

The moment of the ring current of height h and thickness dr equal:

$$dm_x = \pi r^2 dI = \pi r^2 j h dr = -\frac{\pi \dot{B} h}{2\rho} r^3 dr,$$

from which:

$$m_x = -\frac{\pi \dot{B} h}{2\rho} \int_0^{r_0} r^3 dr.$$

Integrating, we find:

Answer:

$$\vec{m} = -\vec{e}_x \cdot \frac{\pi r_0^4 h \dot{B}}{8\rho}.$$

C3

0.50 pts

Determine the magnetic moment $\vec{m}$ of the ball. Express your answer in terms of $\vec{e}_x$, R_0 , ρ and $\dot{B}$.

Let's use spherical coordinates (that is, we will count the angle θ from the positive direction of the axis x). Then for r_0 and h we have:

$$r_0 = R_0 \sin \theta \quad h = d(R_0(1 - \cos \theta)) = R_0 \sin \theta d\theta.$$

Then for element magnetic moment sphere have:

$$dm_x = -\frac{\pi R_0^5 \dot{B}}{8\rho} \sin^5 \theta d\theta,$$

from which:

$$m_x = -\frac{\pi R_0^5 \dot{B}}{8\rho} \int_0^\pi \sin^5 \theta d\theta.$$

Integrate obtain expression with means substitution $x = \cos \theta$:

$$\int_0^\pi \sin^5 \theta d\theta = \int_{-1}^1 (1 - x^2)^2 dx = \int_{-1}^1 (1 - 2x^2 + x^4) dx = \left(x - \frac{2x^3}{3} + \frac{x^5}{5} \right) \Big|_{-1}^1 = \frac{16}{15},$$

from:

Answer:

$$\vec{m} = -\vec{e}_x \cdot \frac{2\pi R_0^5 \dot{B}}{15\rho}.$$

C4

0.40 pts

Obtain the time derivative of the magnetic field induction of the ring at the center of the ball dB_x/dt , which is equivalent to the value $\dot{B}$. Express your answer in terms of v , I , R , x and the magnetic constant μ_0 .

When the ball moves, the magnetic field induction $B_x(x) = B_x(x(t))$, i.e., is a complex function of

time, so we have:

$$\dot{B}_x = \frac{dB_x}{dt} = \frac{dB_x}{dx} \frac{dx}{dt} = v \frac{dB_x}{dx}$$

Find derivative dB_x/dx :

$$\frac{dB_x}{dx} = -\frac{3\mu_0 I R^2 x}{2(R^2 + x^2)^{\frac{5}{2}}},$$

from where:

Answer:

$$\dot{B} = -\frac{3\mu_0 I R^2 x v}{2(R^2 + x^2)^{\frac{5}{2}}}.$$

C5

0.50 pts

Find the proportionality factor $\beta(x)$. Express your answer in terms of I , R , x , R_0 , ρ and the magnetic constant μ_0 .

Since the ball moves along the x axis:

$$\vec{F} = m_x \cdot \frac{\partial \vec{B}}{\partial x} = \vec{e}_x \cdot m_x \frac{dB_x}{dx}.$$

For magnetic moment m_x have:

$$m_x = -\frac{2\pi R_0^5 v}{15\rho} \cdot \frac{dB_x}{dx},$$

from which:

$$F_x = -\frac{2\pi R_0^5}{15\rho} \left(\frac{dB_x}{dx} \right)^2 v.$$

Substitute dB_x/dx , we find:

Answer:

$$\beta(x) = \frac{3\pi\mu_0^2 I^2 R^4 R_0^5 x^2}{10\rho(R^2 + x^2)^5}.$$

C6

0.80 pts

Determine the resistivity ρ of the ball used in the first experiment. Express your answer in terms of m , k , R_0 , R , H , I and the magnetic constant μ_0 .

The first peak is reached at $\Delta x \approx 47$ a.u., and the 12th at $\Delta x \approx 6$ a.u.. Hence:

$$\frac{\gamma}{\sqrt{\omega_0^2 - \gamma^2}} \approx \frac{\ln\left(\frac{47}{6}\right)}{11 \cdot 2\pi} \approx 0,0297 \Rightarrow \frac{\gamma}{\omega_0} \approx 0.03 \ll 1.$$

At the same time have:

$$\frac{\omega_0}{2\gamma} = \frac{\sqrt{mk}}{\beta(H)},$$

from which:

$$\beta(H) \approx \frac{2\gamma\sqrt{mk}}{\omega_0},$$

and finally

Answer:

$$\rho = 15.7 \cdot \frac{\mu_0^2 I^2 R_0^5 R^4 H^2}{\sqrt{mk}(R^2 + H^2)^5}$$

C7

0.70 pts

Determine the resistivity ρ of the ball used in the second experiment. Express your answer in terms of m , k , R_0 , R , H , I and the magnetic constant μ_0 .

From the expression for the resonant amplitude we find:

$$Q = \frac{A_{\text{res}}}{A_0} = \frac{\sqrt{mk}}{\beta(H)} \approx 25$$

from where finally:

Answer:

$$\rho = 23.6 \frac{\mu_0^2 I^2 R_0^5 R^4 H^2}{\sqrt{mk} (R^2 + H^2)^5}$$

D1

0.60 pts

Determine the induction B_z of the magnetic field of the solenoid, as well as its derivative dB_z/dz at the point with coordinate z . Express your answer using μ_0 , n , I , R and z .

From the solid angle theorem for the magnetic field we have:

$$B_z = \frac{\mu_0 i \Omega_{\text{side}}}{4\pi},$$

where $i = In$ – linear density current solenoid, and Ω_{side} – the solid angle subtended by its lateral surface. The solid angle subtended by the lateral surface of the solenoid equals the solid angle that can be seen it base from point with coordinate z , therefore have:

$$\Omega_{\text{side}} = 2\pi(1 - \cos \alpha) = 2\pi \left(1 - \frac{z}{\sqrt{z^2 + R^2}} \right).$$

Thus:

Differentiating, we find:

Answer:

$$B_z = \frac{\mu_0 n I}{2} \left(1 - \frac{z}{\sqrt{z^2 + R^2}} \right).$$

D2

1.00 pts

Determine the linear current density i on the surface of the cylinder at the point with coordinate z . Express your answer using μ_0 , x and $dB_z(z)/dz$.

Since the magnetic field induction outside the cylinder can be considered equal to the magnetic field induction of the solenoid, we have:

$$B_{z(\text{in})} = B_z(z - x) \quad B_{z(\text{out})} = B(z).$$

From the circulation theorem for the magnetic-field induction we obtain:

$$(B_{z(\text{in})} - B_{z(\text{out})}) \approx -x \frac{dB_z}{dz} = \mu_0 i x,$$

from where:

Answer:

$$i(z) = -\frac{x}{\mu_0} \frac{dB_z}{dz}.$$

D3

1.50 pts

Determine the force F_x acting on the cylinder from the magnetic field of the solenoid. Express your answer in terms of μ_0 , r , R , n , I and x .

The magnetic moment of an infinitesimal section of a cylinder is:

$$dm_z = i(z)\pi r^2 dz.$$

For act on it force dF_x have:

$$dF_x = dm_z \frac{dB_z}{dz}.$$

Substitute expression for i , obtain:

$$dF_x = -\frac{\pi r^2 x}{\mu_0} \left(\frac{dB_z}{dz} \right)^2 dz.$$

Use expression for dB_z/dz :

$$dF_x = -\frac{\mu_0 \pi r^2 n^2 I^2 x R^4 dz}{4(R^2 + z^2)^3},$$

from which taking into account the larger distance of the end of the cylinder from the base of the solenoid:

$$F_x \approx -\frac{\mu_0 \pi r^2 n^2 I^2 R^4 x}{4} \int_{-\infty}^{\infty} \frac{dz}{(R^2 + z^2)^3}.$$

Using the substitution $z = R \tan \varphi$ we obtain:

$$F_x = \frac{\mu_0 \pi r^2 n^2 I^2 R^4 x}{4} \int_{-\infty}^{\infty} \frac{dz}{(R^2 + z^2)^3} = \frac{\mu_0 \pi r^2 n^2 I^2 x}{4R} \int_{-\pi/2}^{\pi/2} \cos^4 \varphi d\varphi.$$

Let us compute last integral:

$$\begin{aligned} \int_{-\pi/2}^{\pi/2} \cos^4 \varphi d\varphi &= \int_{-\pi/2}^{\pi/2} \left(\frac{1 + \cos 2\varphi}{2} \right)^2 d\varphi = \int_{-\pi/2}^{\pi/2} \left(\frac{1}{4} + \frac{\cos 2\varphi}{2} + \frac{\cos^2 2\varphi}{4} \right) d\varphi = \\ &= \int_{-\pi/2}^{\pi/2} \left(\frac{1}{4} + \frac{\cos 2\varphi}{2} + \frac{1}{4} \left(\frac{1 + \cos 4\varphi}{2} \right) \right) d\varphi = \int_{-\pi/2}^{\pi/2} \left(\frac{3}{8} + \frac{\cos 2\varphi}{2} + \frac{\cos 4\varphi}{8} \right) d\varphi = \\ &= \left(\frac{3\varphi}{8} + \frac{\sin 2\varphi}{4} + \frac{\sin 4\varphi}{32} \right) \Big|_{-\pi/2}^{\pi/2} = \frac{3\pi}{8}. \end{aligned}$$

So way:

Answer:

$$F_x = -\frac{3\pi^2 \mu_0 \pi r^2 n^2 I^2}{32R} x.$$

D4

0.30 pts

Obtain the dependence of the displacement of the rod x on time t . Express your answer in terms of μ_0 , r , R , n , I and m .

Let us write the equation of motion of the cylinder:

$$m\ddot{x} = -\frac{3\mu_0\pi^2 r^2 n^2 I^2}{32R}x \Rightarrow \omega_0^2 = \frac{3\mu_0\pi^2 r^2 n^2 I^2}{32mR} \Rightarrow x(t) = A \sin \omega_0 t + B \cos \omega_0 t.$$

Determinable constant A and B from initial condition:

$$\begin{cases} x(0) = B = 0 \\ v_x(0) = \omega_0 A = v_0 \end{cases} \Rightarrow A = \frac{v_0}{\omega_0}$$

Thus:

Answer:

$$x(t) = v_0 \sqrt{\frac{32mR}{3\mu_0\pi^2 r^2 n^2 I^2}} \sin \sqrt{\frac{3\mu_0\pi^2 r^2 n^2 I^2}{32mR}} t.$$